Kick-off Diagnostic · L9-2 — Circles

Grade 11

8 questions · 30 seconds each · about 4 minutes · Diagnose phase (6 min of 60)

Pre-session anchor (stations 1 and 2 of the pre-session say this, so items 2 and 6 test it).
The one idea: A circle is every point at distance r from its center, so $(x-h)^2+(y-k)^2=r^2$ has center (h,k) and radius r; completing the square rewrites any circle equation into this form.
The mistake most students make: Reading $r^2=25$ as $r=25$, or copying the signs of h and k from the brackets.
Readiness items 3, 4, 5, 7, 8 test: the distance formula, completing the square, and solving a quadratic.
Wayground set-up: timer 30 s on every question; shuffle options on, questions off; hide the leaderboard; show answers after the quiz, not after each item. Item 1 is a poll (no points). Items 2 to 8 are 1 point each. Tags below: **maroon** = tells you about the pre-session, **navy** = readiness for today. **Green** = correct.

1 POLL · 30 s — SELF-REPORT

Be honest: how much of the pre-session task did you do?

A. All of it, including the readiness check
B. I watched and read but skipped the check
C. I opened it but did not finish
D. I did not open it

Signal. Ungraded. Tally it first; it is the self-report half of the pre-session test.

2 MULTIPLE CHOICE · 30 s — PRE-SESSION CHECK

In the pre-session, $x^2+y^2-6x+4y-12=0$ became a circle with right-hand side 25. What is the radius?

✓ **A.** 5
B. 25
C. 12
D. $\sqrt{12}$

Signal. Wrong → the pre-session mistake. B = read r^2 as r; C = the constant of the original; D = square-rooted the wrong number.

3 FILL IN THE BLANK · 30 s — READINESS

The distance between (1, 2) and (4, 6) is _____

Accept: 5

Signal. 25 = forgot the square root; 7 = added the differences. The distance formula is the circle's definition.

4 CHECKBOX · 30 s — READINESS

Select the **two** correct completions of the square.

✓ **A.** $x^2-10x=(x-5)^2-25$
✓ **B.** $x^2+4x=(x+2)^2-4$
C. $x^2-6x=(x-3)^2-6$
D. $x^2+8x=(x+8)^2-64$

Signal. C = subtracted 6 instead of 9; D = used the whole coefficient, not half.

5 MATCH · 30 s — READINESS

Match each circle with its center.

$(x-1)^2+(y-4)^2=9$ → (1, 4)
$(x+1)^2+(y+4)^2=9$ → $(-1,-4)$ (shown shuffled in Wayground)
$(x-4)^2+(y+1)^2=9$ → $(4,-1)$

Signal. Mixing the first and third = swapped h and k; sign errors = copied the brackets.

6 MULTIPLE CHOICE · 30 s — PRE-SESSION CHECK

What are the center and radius of $(x+2)^2+(y-3)^2=49$?

✓ **A. center $(-2, 3)$, radius 7**
B. center $(2, -3)$, radius 7
C. center $(-2, 3)$, radius 49
D. center $(2, -3)$, radius 49

Signal. Pre-session item. B = copied the signs from the brackets; C = read r^2 as r; D = both errors.

7 REORDER · 30 s **READINESS**

Put the steps in order to solve $x^2 + y^2 = 25$ with $y = x + 1$.

Correct order: substitute: $x^2 + (x+1)^2 = 25$ → collect: $2x^2 + 2x - 24 = 0$ → divide by 2 and factor: $(x+4)(x-3) = 0$ → use $y = x + 1$: $(-4, -3)$ and $(3, 4)$

Signal. Wrong → substituting a line into a circle is the new move today; check the order of expand, collect, factor.

8 DROPDOWN · 30 s **READINESS**

Solve $x^2 + x - 12 = 0$.

✓ **A.** $x = 3$ **or** $x = -4$

B. $x = -3$ or $x = 4$

C. $x = 12$ or $x = -1$

D. $x = 3$ or $x = 4$

Signal. B = signs reversed; C = factor pair product -12 but sum 11; D = both positive.

Read the results (90 seconds)

Readiness R — **Items 3, 4, 5, 7, 8 (out of 5). 0–2** → re-teach the distance formula and completing the square with me. **3–4** → straight to deriving the circle equation and line-circle systems; fix the missed item in passing. **5** → start the Investigate prompt now.

Pre-session P — **Items 2 and 6 (out of 2), read with the item 1 poll.**

What you see	Reading	What you do
Poll "all" or "read", $P = 2$	Prepared	Nothing extra. Use their R score to route.
Poll "all" or "read", $P = 0$	Claimed, did not retain	Two-minute recap of the one idea at the start. Ask them to restate it in their own words.
Poll "opened" or "did not open"	Task not done	No penalty. Pair with a prepared partner for Targeted Instruction; note it for your task tracker.
$R \leq 2$ and $P = 2$	Content gap	Re-teach group. The task was done, so the idea needs a different explanation.
$R \geq 4$ and $P \leq 1$	Task skipped, skills fine	Move on. Chase the task, not the content.

Class-level rule. If more than 30% of the class chose "opened" or "did not open", or the class average on P is below 1, spend three minutes recapping the one idea above before Targeted Instruction. If more than 30% score $R \leq 2$, the whole class takes the re-teach route first.